

Open Source — Detailed Lecture Notes

JKSSB Computer Awareness

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45.1 Source code and what it means

Every program begins as **source code**: the set of instructions a programmer writes in a programming language such as C, Java, or Python. Source code is *human-readable* — a person can open the file, read it, and change it. Before the computer can run the program, a **compiler** or an **interpreter** converts the source code into a machine-readable form. That machine-readable form is called **object code**, or simply the executable, and it is what the processor actually executes. Ordinary users of a finished program normally receive only this compiled form and never see the source code.

Definition (Source code). *The human-readable set of instructions, written in a programming language, that defines what a program does; it is converted into machine-readable object code by a compiler or an interpreter.*

The important consequence is that whoever holds the source code controls how the software can be studied, changed, and shared. The whole topic of open source versus proprietary software grows out of this single fact.

45.2 Open source versus proprietary software

Software falls into two broad camps according to what is done with its source code.

Point	Open source	Proprietary (closed source)
Source code	Publicly available	Kept secret
View or modify	Allowed, subject to the licence	Not allowed without permission
Redistribute	Allowed, subject to the licence	Not allowed
Typical user gets	Source code plus executable	Only the compiled executable

Open-source software makes its source code public. Anyone is allowed to inspect it, modify it, and redistribute it, subject to the terms of its licence. **Proprietary software** keeps its source code confidential; the user receives only the compiled executable and may not view, change, or share the source without the owner's permission. Examples of proprietary software include Microsoft Windows, Apple macOS, Microsoft Office, Adobe Photoshop, and Oracle Database. The deciding factor between the two is the source code, not the price: some open-source software is sold, and some proprietary software is given away free of cost.

Definition (Open source). *A software development model in which the source code is made publicly available so that anyone may view, modify, and redistribute it under the terms of its licence.*

45.3 The open-source idea: view, modify, share

The meaning that examiners most often test is the three-part permission of an open-source licence. First, a user may **view** the source code and see exactly how the software works. Second, a user may **modify** it — fix a bug, improve it, or add a new feature. Third, a user may **share** it, redistributing copies and modified versions to others. These permissions are granted by the software’s licence, so the exact conditions differ from one project to another; but the underlying spirit is the same three freedoms in the code.

Example 45.3.1. *An item states, “An open-source licence allows a user to view, change, and share the source code.” This is the standard, correct description and is the answer expected in the exam. An option that says only “software available free of cost” describes freeware, not open source.*

45.4 Freeware, free software, public domain, commercial, shareware

Several labels are often mixed up with open source. Each has a distinct meaning.

Term	What is free	Source code
Open source	Use, study, change, share	Available
Freeware	Cost only	Usually not provided
Free software	The user’s freedom	Available
Shareware	A limited trial period	Not provided
Public domain	No copyright at all	No restrictions
Commercial	Nothing (it is sold)	Usually not provided

Freeware is software given away free of cost, but its source code is usually not provided. **Free software** is about freedom, not price: the user is free to study, change, and share it. **Public domain** software has no copyright owner, so anyone may use, modify, and redistribute it without any restriction. **Shareware** is distributed as a free trial, working for a limited period or with limited features; to keep using it, the user must pay. **Commercial software** is developed and sold for profit and is often, though not always, proprietary.

Definition (Freeware). *Software distributed free of cost, whose source code is usually not made available; it is free in price, not in freedom.*

Definition (Shareware). *Software distributed as a trial version, free for a limited time or with limited features, after which payment is required to continue using it.*

45.5 Free software and its four freedoms

The term **free software** comes from the **Free Software Foundation (FSF)**, founded by **Richard Stallman**. In this context, “free” refers to liberty, not price — the well-known phrase is “free as in freedom, not free as in free beer”. The FSF defines four freedoms: the freedom to run the program for any purpose; the freedom to study how it works and change it; the freedom to redistribute copies to help others; and the freedom to distribute modified versions to the public. Free software and open source are closely related and frequently overlap, but they are not identical labels. Free software stresses the ethical freedom of the user; open source stresses the practical benefits of a publicly available source code.

Definition (Free software). *Software that grants the user the freedom to run, study, change, and redistribute it and its modified versions; “free” here means freedom, not zero cost.*

45.6 The Open Source Initiative

Alongside the Free Software Foundation stands the **Open Source Initiative (OSI)**, founded in 1998. The OSI maintains the **Open Source Definition**, the set of criteria a licence must satisfy to be called an open-source licence, and it reviews and approves licences against those criteria. The term **open source** itself was popularised in 1998 as a more business-friendly label than “free software”. The two movements overlap heavily: most licences approved by the OSI are also free-software licences. An umbrella term often seen in questions is **FOSS**, meaning Free and Open-Source Software, or **FLOSS** (Free/Libre and Open-Source Software), which covers both traditions together.

The Open Source Definition lays down the conditions a licence must meet. The software must be freely redistributable; the source code must be included or easily obtainable; derived works and modifications must be allowed; the licence must not discriminate against any person, group, or field of use; and the rights must apply to everyone who receives the software. A licence that satisfies these conditions is called an **OSI-approved** licence. This is why “open source” is a matter of the licence, not simply of whether the software is free of cost.

Example 45.6.1. *Two programs are both free to download. Program A ships with its source code under an OSI-approved licence that permits modification and redistribution; Program B ships only compiled files and forbids copying. A is open source; B is freeware. The download price is identical, but the licences differ, and the licence is what decides the category.*

45.7 Benefits and limitations of open source

Open source is chosen for several practical reasons. The licence cost is usually zero, so organisations avoid per-seat fees. The source code can be inspected, which lets many people review it for bugs and security weaknesses. Users are not locked in to a single vendor, and the software can be customised to local needs. A large community often provides documentation and support.

Open source also has limitations that a fair answer should acknowledge. A popular project may have no formal commercial support contract, so support depends on the community or on paid third-party vendors. Compatibility with proprietary formats and integration with existing systems can be difficult. The total cost of ownership is not always zero, because training, migration, and maintenance still cost money. These points are occasionally tested as “which of the following is a limitation of open source”.

45.8 Full forms and key names to remember

Direct-recall items often ask for an expansion or for the name behind a term. The following are the ones most worth memorising.

Short form	Full form / meaning
GPL	GNU General Public License
FSF	Free Software Foundation
OSI	Open Source Initiative
FOSS	Free and Open-Source Software
FLOSS	Free/Libre and Open-Source Software
AOSP	Android Open Source Project
MIT	MIT License (a permissive software licence)
BSD	Berkeley Software Distribution (its licence family)

Remember too the names behind the ideas: the Free Software Foundation was founded by **Richard Stallman**, and **Git** was created by Linus Torvalds to manage the Linux kernel’s source code. **GitHub**, **GitLab**, **Bitbucket**, and **SourceForge** are the best-known hosting sites for open-source repositories.

45.9 Repositories and community development

Open-source projects live in a **repository** (often shortened to “repo”). A repository is a storage location that holds a project’s source code together with every change ever made to it. **Version control** records this history, so an earlier version can be restored if needed. **Git** is the most widely used distributed version-control system, and popular hosting sites for repositories include **GitHub**, **GitLab**, **Bitbucket**, and **SourceForge**.

Open-source software is usually built by a **community** of contributors spread around the world. A contributor may **fork** a project, that is, make a personal copy for experimenting or improving. A **pull request** is then a proposal to merge those changes back into the main project, where **maintainers** review them. An **issue tracker** records bugs and feature requests. Because many people can inspect and improve the code, community review is treated as a key strength of open-source development.

45.10 Basic licensing concepts

By default, **copyright** gives the creator of a work exclusive rights over it. A **licence** is the legal permission that states what others may do with the software. Open-source and free-software licences fall into two broad families.

Copyleft licences, such as the GNU General Public License (GPL), require that modified versions of the software be released under the *same* licence. This keeps the software and its derivatives open. **Permissive** licences, such as the MIT License, the Apache License 2.0, and the BSD License, allow the code to be reused with very few conditions, including inside proprietary projects, provided the licence notice is kept. The licence, not the price, decides whether software is truly open source.

Definition (Copyleft). *A licence condition that requires any modified or extended version of the software to be distributed under the same licence, so the work cannot be closed.*

45.11 Copyright, licences, and piracy

It helps to separate three ideas. **Copyright** is the creator’s legal ownership of the work. A **licence** is the permission the owner grants for specific uses. **Piracy** is the use, copying, or distribution of software in a way the licence forbids — for example, installing a proprietary program without a valid licence, or copying it to machines the licence does not cover. Open-source software is not “unlicensed”: it is licensed, and its licence grants broad permissions. Public-domain software, by contrast, has no owner and therefore needs no licence at all. The exam sometimes tests this by asking whether open-source software is “free of copyright”; the answer is no, because the copyright remains with the authors and the licence grants the freedoms.

Example 45.11.1. *A student installs the same licensed copy of a proprietary office suite on five computers when the licence permits only one. This is **piracy**. Installing an open-source office suite such as LibreOffice on all five computers, in line with its licence, is not piracy. The difference is not the price paid but whether the use matches what the licence allows.*

45.12 Common licences at a glance

Licence	Type	Main condition
GNU GPL	Copyleft	Derivative works must use the same licence
MIT License	Permissive	Reuse allowed; keep the licence notice
Apache License 2.0	Permissive	Few limits, plus an explicit patent grant
BSD License	Permissive	Very few conditions
Creative Commons	Content licence	For creative works, not usually software

Note that **Creative Commons** licences are designed for creative works such as text, images, and music; they are not typically used to license software. For software, the GPL, MIT, Apache, and BSD licences are the ones to recognise.

45.13 Well-known open-source and proprietary software

Familiar examples help fix the concepts.

Open source	Proprietary
Linux (operating system)	Microsoft Windows
Android (mobile operating system)	Apple macOS, iOS
Mozilla Firefox (browser)	Microsoft Office
Chromium (Chrome’s open-source base)	Adobe Photoshop
LibreOffice (office suite)	Oracle Database
VLC media player, GIMP, Blender	
Apache HTTP Server, MySQL, Python, Git, WordPress	

Android is the open-source **mobile operating system**, built on the open-source Linux kernel. This is a directly tested fact: when a question asks for the open-source mobile OS, the answer is Android. In contrast, iOS is a proprietary mobile operating system, and Windows and macOS are proprietary desktop operating systems.

45.14 Exam retrieval and common confusions

Five distinctions prevent most errors on this topic.

1. The test of open source is the **source code**, not the price. An open-source licence lets users **view, change, and share** the source (PYQ-26).
2. **Freeware** is free of **cost**; **free software** is about **freedom**. Do not treat the two as the same (TRAP-23).
3. **Open source**, **freeware**, and **shareware** are three different things: one shares the source, one is free of cost, one is a paid trial (TRAP-23).
4. The open-source **mobile operating system** is **Android** (PYQ-24); iOS is proprietary.
5. **GPL** is copyleft (derivatives stay open); **MIT**, **Apache 2.0**, and **BSD** are permissive.

Example 45.14.1. *An item asks, “Which of the following is an open-source mobile operating system?” Trace the options: Android is open source (built on the Linux kernel), while iOS, Windows, and macOS are proprietary. The answer is Android. If the stem instead asks what an open-source licence permits, the answer is the right to view, change, and share the source code.*

Revision at a glance

Term	What it is	What it is not
Open source	Source code is public and licensed for view/change/share	Not merely free of cost
Proprietary	Source kept secret; only the executable is distributed	Not necessarily paid
Freeware	Free of cost	Not necessarily open source
Free software	Freedom to run, study, change, share	Not about zero price
Public domain	No copyright owner	Not the same as open source
Shareware	Free trial, then pay	Not free forever
Commercial	Sold for profit	Not necessarily proprietary or open
Copyleft (GPL)	Derivatives must stay open	Not permissive
Permissive (MIT)	Reuse with few conditions	Not copyleft

Exercises

1. Define source code and explain how it differs from object code.
2. State the three things an open-source licence typically permits a user to do with the source code.
3. Distinguish freeware from free software.
4. Explain what public domain software is and how it differs from open-source software.
5. Name one copyleft licence and one permissive licence, and state the main condition of each.

Solutions

1. Source code is the human-readable set of instructions written by a programmer in a programming language. It is converted by a compiler or interpreter into object code, the machine-readable form the computer actually runs.
2. View the source code, modify it, and share (redistribute) it, subject to the licence terms.
3. Freeware is free of cost but its source code is usually not provided. Free software is about freedom: the user may study, change, and share the software; “free” means liberty, not price.
4. Public domain software has no copyright owner and carries no restrictions at all. Open-source software is still copyrighted, but its licence grants the freedom to view, change, and share the source.
5. Copyleft: the GNU GPL, which requires modified versions to be released under the same licence. Permissive: the MIT License (or Apache 2.0 / BSD), which allows reuse with few conditions, such as keeping the licence notice.